

Configuration Automation using Ansible

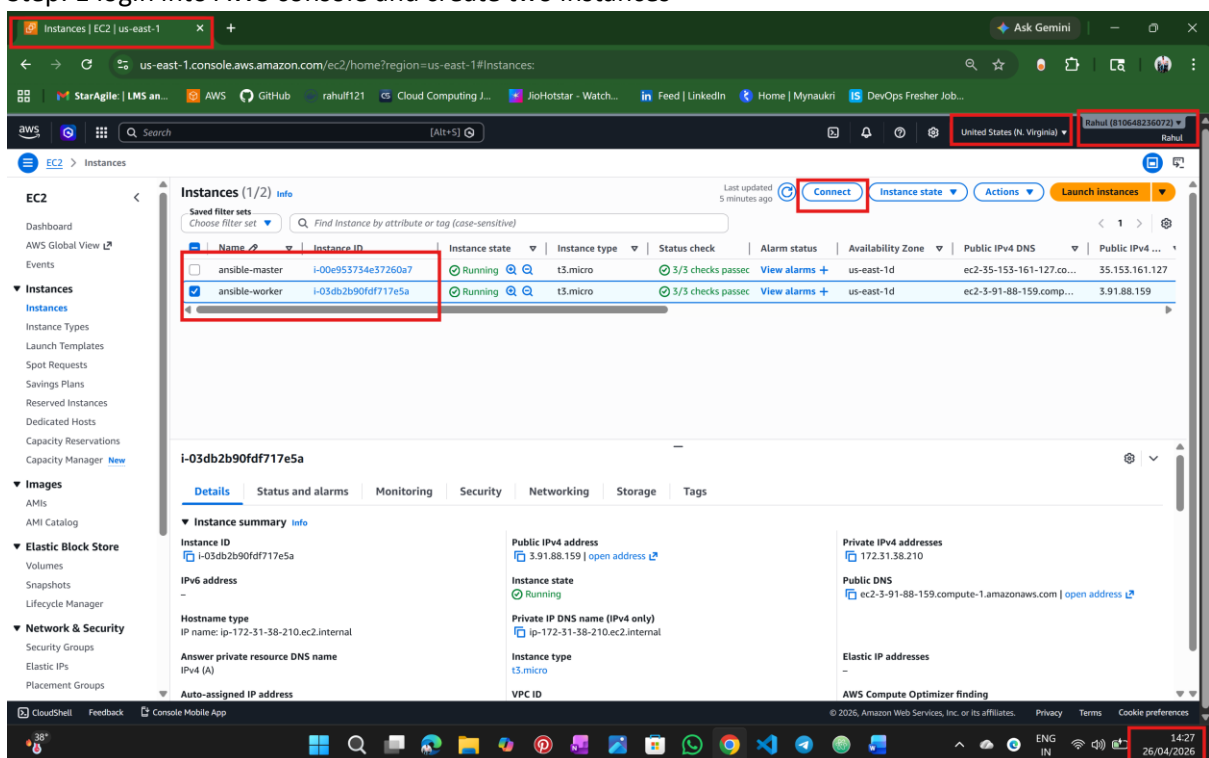
Submitted to - Vikul Sir

Date of Submission – 28/APR/2026

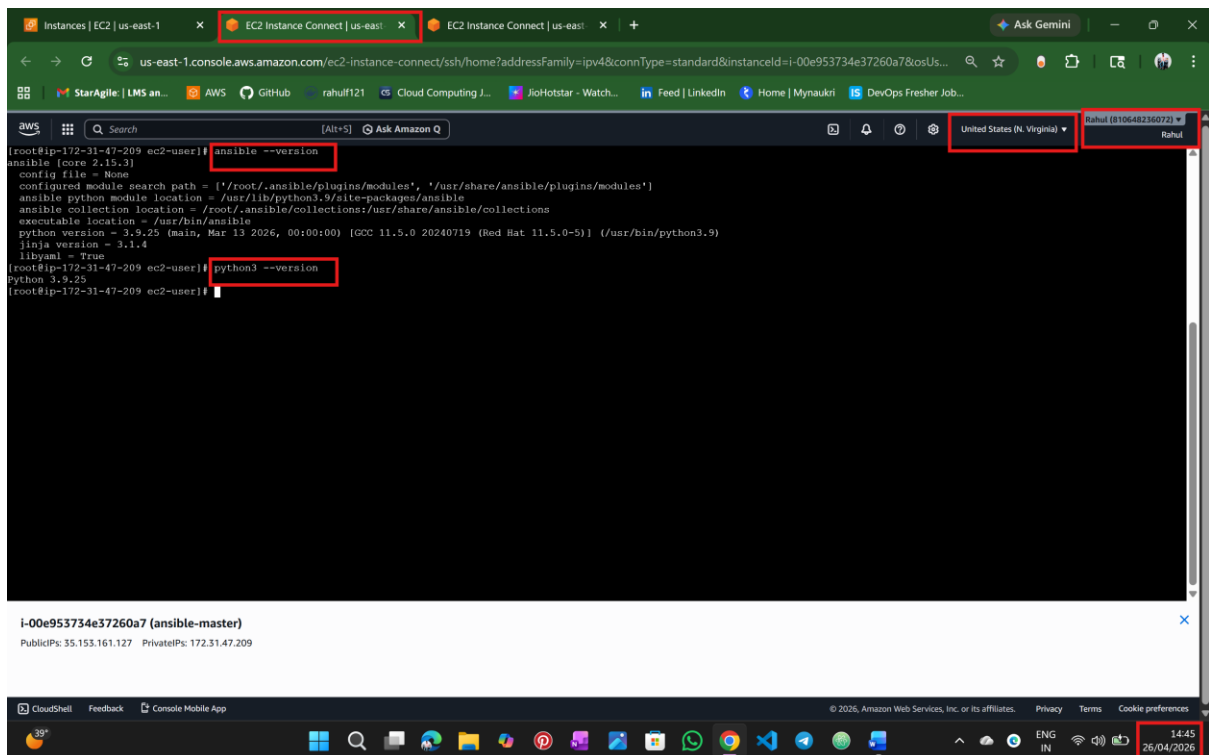
Submitted by – Rahul Nalathathi

L1 - Create and Execute Ansible Playbook to Setup Java Maven Application Build Server

Step: 1 login into AWS console and create two instances



Step: 2 Master node Terminal

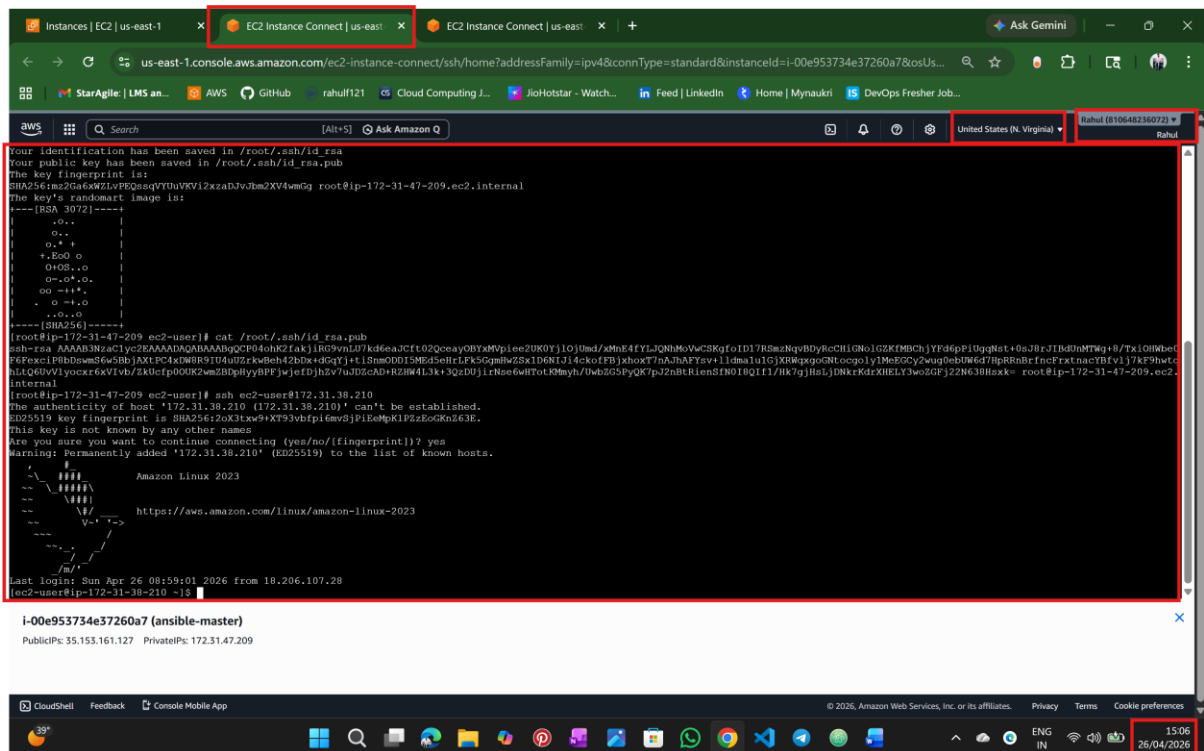


The screenshot shows the AWS CloudShell interface for an EC2 instance named 'i-00e953734e37260a7' (ansible-master). The terminal window displays the following commands and output:

```
[root@ip-172-31-47-209 ec2-user]# ansible --version
ansible [core 2.15.3]
  config file = None
  configured module search path = ['/root/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3.9/site-packages/ansible
  executable location = /usr/bin/ansible
  python version = 3.9.25 (main, Mar 13 2026, 00:00:00) [GCC 11.5.0 20240719 (Red Hat 11.5.0-5)] (/usr/bin/python3.9)
  jinja version = 3.1.4
  libyaml = True
[root@ip-172-31-47-209 ec2-user]# python3 --version
Python 3.9.25
[root@ip-172-31-47-209 ec2-user]#
```

The interface includes a top navigation bar with tabs for 'Instances | EC2 | us-east-1', 'EC2 Instance Connect | us-east-1', and 'EC2 Instance Connect | us-east-1'. The bottom status bar shows the time as 14:45 on 26/04/2026.

Step:3 connect the master node and worker node with ssh-key

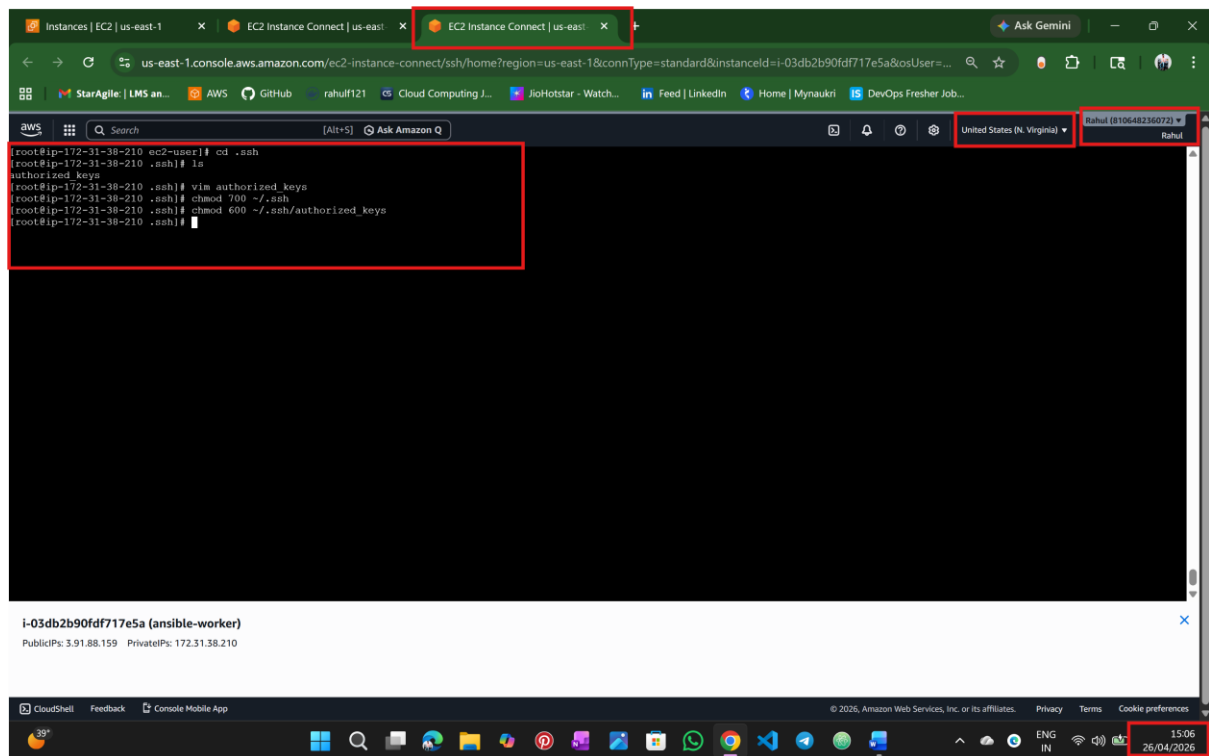


The screenshot shows the AWS CloudShell interface for the same EC2 instance. The terminal window displays the following commands and output:

```
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:zG4xwz1wPZQaQYUUVKXV12xzdJvJhm2XV4wGg root@ip-172-31-47-209.ec2.internal
The key's randomart image is:
[SHA256]
+-----+
|o..+|
|o..+|
|o..+|
|+Eoo+|
|OHO..o|
|o..o..o|
|oo..+..+|
|o..+..+|
|..o..+..+|
|..o..+..+|
+-----+
[root@ip-172-31-47-209 ec2-user]# cat /root/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCAk4k1R69vndJ7kd6eAJCt02Qceay0BYxMplie2UK0Yj10Jwd/xMnE4fYLJQNMb0PNC5KgoID17RSmzNqyB0DyRcChlGNoLgZKfMBChjYfdePPlUggNet4ba38rJIBdUnMTWg8/Tx1OHwBelPFExc1P8BDeam6Ew5Bb)AX1PC4xkMB891U4u7zrkWBeh42bDx+dgGy)++tISmODD1SME45Hr1FK5Ggmw25x1D6N1J14kofFBj)khoxT7nA3hAFtsv+ldma1u1GjXRWqgoQNTocpolylMeGCy2wug0ebUW6d7BpRRnBfneFrtxnacTB1v17KF9hwtcBlt0G0V1yocxrKv5vb/ZK0cp000R2m2B8pvyBPFjweJDjhZv7uJDZcAD+RZHW4L3k+3QzD0JiRNae6WHTotKRMmYh/UwbZG5PyQ7KpJ2nBtRlenSfN01BQ1f1/Hk7gHsljDNkrKdxXHELY3woGFj22N638Hsxk- root@ip-172-31-47-209.ec2.internal
[root@ip-172-31-47-209 ec2-user]# ssh ec2-user@172.31.38.210
The authenticity of host '172.31.38.210 (172.31.38.210)' can't be established.
ED25519 key fingerprint is SHA256:26X3tXw9XT93vbfpi6v8S)PIEeMpK1P2zEoGKnZ63E.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.31.38.210' (ED25519) to the list of known hosts.
Amazon Linux 2023
--
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Sun Apr 26 08:59:01 2026 from 18.206.107.28
ec2-user@ip-172-31-38-210 ~$
```

The interface includes the same top navigation bar. The bottom status bar shows the time as 15:06 on 26/04/2026.

Step:4 this is worker node terminal paste the ssh-key in worker node

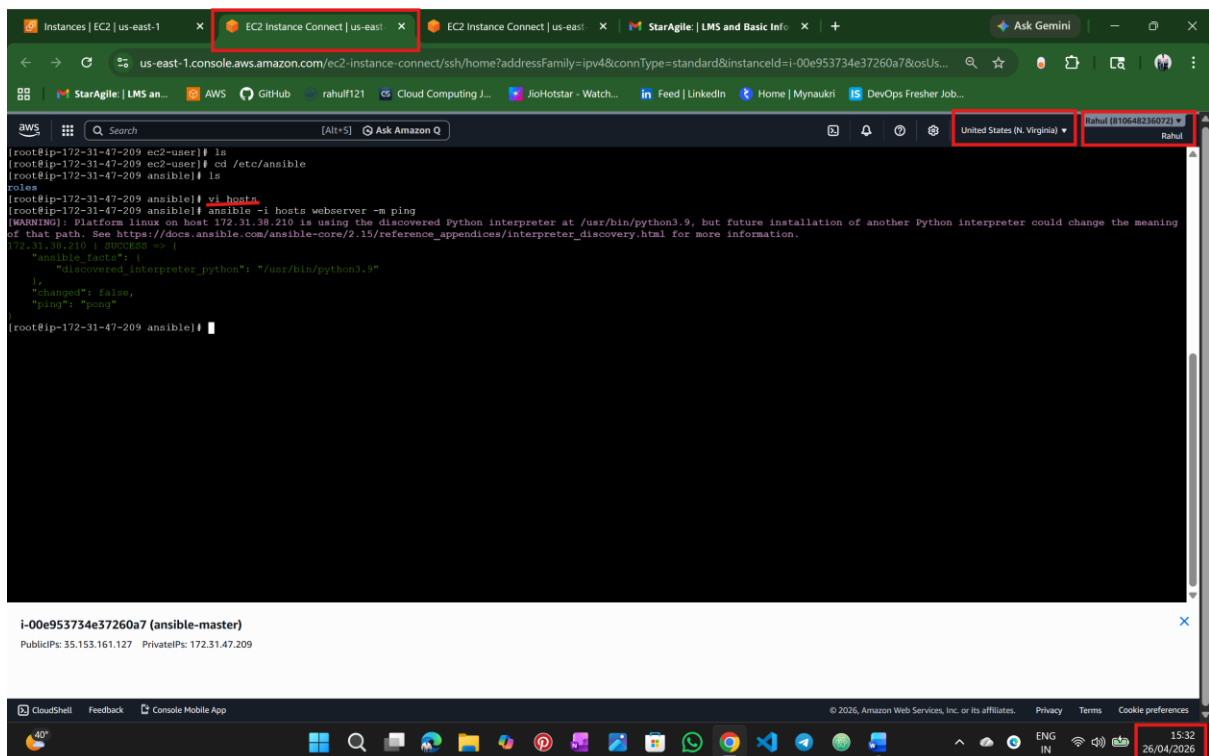


The screenshot shows the AWS CloudShell interface for an EC2 instance named 'i-03db2b90fd717e5a (ansible-worker)'. The terminal window displays the following commands and output:

```
[root@ip-172-31-38-210 ec2-user]# cd .ssh
[root@ip-172-31-38-210 .ssh]# ls
authorized_keys
[root@ip-172-31-38-210 .ssh]# vim authorized_keys
[root@ip-172-31-38-210 .ssh]# chmod 700 ~/.ssh
[root@ip-172-31-38-210 .ssh]# chmod 600 ~/.ssh/authorized_keys
[root@ip-172-31-38-210 .ssh]#
```

The terminal output shows the instance's public and private IP addresses: PublicIPs: 3.91.88.159 PrivateIPs: 172.31.38.210. The bottom status bar indicates the time is 15:06 on 26/04/2026.

Step:5 verify whether master node and worker node is connected are not? Yes! It successfully connected.



The screenshot shows the AWS CloudShell interface for an EC2 instance named 'i-00e953734e37260a7 (ansible-master)'. The terminal window displays the following commands and output:

```
[root@ip-172-31-47-209 ec2-user]# is
[...]
```

The terminal output shows the instance's public and private IP addresses: PublicIPs: 35.153.161.127 PrivateIPs: 172.31.47.209. The bottom status bar indicates the time is 15:32 on 26/04/2026.

Step:6 Now, write the playbook yaml file to install java and maven

The screenshot shows the AWS CloudShell interface. The terminal displays the following Ansible playbook content:

```
--
- name: Setup Java Maven Build Server
  hosts: webserver
  become: yes

  tasks:
    - name: Install Java
      yum:
        name: java-17-amazon-corretto
        state: present
    - name: Install Maven
      yum:
        name: maven
        state: present
```

The interface includes a top navigation bar with tabs for 'Instances | EC2 | us-east-1', 'EC2 Instance Connect | us-east-1', and 'StarAgile: LMS and Basic Info'. The right sidebar shows the 'United States (N. Virginia)' region and the user 'Rahul (810648236072)'. The bottom status bar indicates the instance 'i-00e953734e37260a7 (ansible-master)' with public IP 35.153.161.127 and private IP 172.31.47.209. The system clock shows 15:37 on 26/04/2026.

Step:7 Run the playbook yaml file

The screenshot shows the AWS CloudShell interface with the Ansible playbook being executed. The terminal output is as follows:

```
[root@ip-172-31-47-209 ansible] vim playbook.yml
[root@ip-172-31-47-209 ansible] ansible-playbook -i hosts playbook.yml

PLAY [Setup Java Maven Build Server] *****

TASK [Gathering Facts] *****
[WARNING]: Platform linux on host 172.31.38.210 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.38.210]

TASK [Install Java] *****
changed: [172.31.38.210]

TASK [Install Maven] *****
ok: [172.31.38.210]

PLAY RECAP *****
172.31.38.210      : ok=3    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

[root@ip-172-31-47-209 ansible] █
```

The interface is identical to the previous screenshot, showing the same navigation tabs, sidebar, and bottom status bar with the instance 'i-00e953734e37260a7 (ansible-master)' and system clock at 15:40 on 26/04/2026.

Step:8 Result, check and verify,

The screenshot shows a web browser window with the AWS console. The active tab is 'EC2 Instance Connect | us-east-1'. The address bar shows the URL: `us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-03db2b90df717e5a&osUser=...`. The browser's address bar and the terminal window title bar are highlighted with red boxes. The terminal window title is 'United States (N. Virginia)'. The terminal output shows the following commands and results:

```
[root@ip-172-31-38-210 ec2-user]# cd .ssh
[root@ip-172-31-38-210 .ssh]# ls
authorized_keys
[root@ip-172-31-38-210 .ssh]# vim authorized_keys
[root@ip-172-31-38-210 .ssh]# chmod 700 ~/.ssh
[root@ip-172-31-38-210 .ssh]# chmod 600 ~/.ssh/authorized_keys
[root@ip-172-31-38-210 .ssh]#
[root@ip-172-31-38-210 .ssh]#
[root@ip-172-31-38-210 .ssh]# java -version
openjdk version "17.0.18" 2024-07-20 LTS
OpenJDK Runtime Environment Corretto-17.0.18.9.1 (build 17.0.18+9-LTS)
OpenJDK 64-Bit Server VM Corretto-17.0.18.9.1 (build 17.0.18+9-LTS, mixed mode, sharing)
[root@ip-172-31-38-210 .ssh]# mvn -version
Apache Maven 3.8.4 (Red Hat 3.8.4-3.amzn2023.0.5)
Maven home: /usr/share/maven
Java version: 17.0.18, vendor: Amazon.com Inc., runtime: /usr/lib/jvm/java-17-amazon-corretto.x86_64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.1.166-197.305.amzn2023.x86_64", arch: "amd64", family: "unix"
[root@ip-172-31-38-210 .ssh]#
```

Below the terminal output, the instance details are shown: `i-03db2b90df717e5a (ansible-worker)`. The public IP is `3.91.88.159` and the private IP is `172.31.38.210`. The bottom of the screenshot shows the Windows taskbar with the date and time `15:43 26/04/2026` highlighted with a red box.

Nalathati Rahul